

On the Nature of Time, Space, Scale, and the Boundaries of Physics

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Abstract

Standard cosmology treats time, space, and scale as properties of a Lorentzian manifold on which physical processes occur. The Holographic Λ CDM (H- Λ CDM) framework inverts this: cosmological structure is the algebraic-modular projection of information processing on the cosmic causal diamond, with the manifold and its observables emergent from the modular state on the algebra of horizon observables $\mathcal{A}(\mathcal{D})$. This paper develops the structural commitments of that inversion. We argue that time, properly understood within the framework, is the rate of state precipitation generated by modular flow at $\gamma(z)$ — a rate that varies across cosmic history and is undetectable to embedded observers because their measuring instruments are themselves slaved to it. We argue that scale, properly understood, is intrinsically cycle-local: there is no cross-cycle absolute meter, no privileged epoch’s second, and no measurement frame from which the universe’s “size” is meaningful in any cycle-independent sense. We argue that the QTEP partition between coherent and decoherent entropy structurally drives cyclicity at the $H \rightarrow 0$ asymptote, with internal observers experiencing cycle transitions as immediate regardless of external duration. We argue that the boundaries of physics, accordingly, are not where measurement precision runs out but where the algebraic structure ceases to support cross-frame comparison. The Hubble tension, the cosmological-constant scale, and the BAO sound horizon are within-cycle observables that probe γ -evolution and hence reveal the framework’s structural commitments empirically; cross-cycle scales are not observables in any meaningful sense. We close by examining what these commitments imply for traditional cosmological puzzles — the size of the universe at the big bang, the question of what came before, the apparent fine-tuning of initial conditions — each of which dissolves or reframes when measurement is treated as intrinsically embedded rather than as access to absolute structure.

Keywords: holographic cosmology; modular flow; relational time; causal diamond; cyclic cosmology; foundations of cosmology

1 The relational structure of cosmological observation

Cosmology has accumulated a series of measurement tensions over the last decade that resist resolution within standard Λ CDM. The Hubble tension between local distance-ladder measurements ($\sim 73 \text{ km s}^{-1} \text{ Mpc}^{-1}$) and CMB-inferred values ($\sim 67 \text{ km s}^{-1} \text{ Mpc}^{-1}$) has reached $\sim 5\sigma$ across multiple independent methodologies [1,2,3]. The DESI 2024 BAO analyses prefer evolving dark energy at $2.6\text{--}3.9\sigma$ [4,5]. The S_8 tension between weak-lensing and CMB-derived structure amplitudes persists at $\sim 2\text{--}3\sigma$ [6]. None of these tensions has yielded to standard parameter-fitting within Λ CDM, despite a decade of focused effort by multiple collaborations and theoretical groups.

The conventional response treats each tension as a problem requiring either new physics added to Λ CDM (early dark energy, modified neutrino sectors, varying fine-structure constant, modified gravity) or unrecognized systematics in one or more measurement chains. Both responses

share an unstated premise: that cosmological observables are properties of a single underlying Lorentzian spacetime against which all measurements should agree, and disagreement signals either physical novelty or measurement error. The premise is so deeply embedded in the standard formalism that it is rarely examined directly.

This paper examines that premise. We argue, following the structural commitments of Λ CDM developed in [7,8,9], that cosmological observables are not measurements of properties of a fixed underlying manifold but are projections of modular-state structure onto the local Lorentzian representation of an embedded observer. The disagreement between local and CMB Hubble measurements is not a tension to be resolved within the standard formalism; it is the structural signature of γ — the modular flow rate on the cosmic causal diamond’s algebra of observables — varying across the cosmic history that separates the two measurement scales. Within Λ CDM, the Hubble tension is not a problem; it is a probe.

We take the resolution of the Hubble tension by Λ CDM [8] as the empirical anchor for the structural argument that follows. The framework’s parameter-free predictions match conservative-TRGB measurements at NGC 4258 ($69.15 \text{ km s}^{-1} \text{ Mpc}^{-1}$) and LMC ($70.40 \text{ km s}^{-1} \text{ Mpc}^{-1}$) anchors across four supernova photometric systems, with cross-anchor structure confirming the predicted logarithmic distance dependence [8]. The cosmological-constant scale follows from the same modular foundation as the inverse holographic information capacity at $z = 0$ [9]. The BAO sound horizon at recombination (150.71 Mpc) emerges from the same construction [7]. Three independent cosmological observables, three quantitative successes from a single foundation. The conceptual paper that follows develops the foundation those successes share.

The rest of this paper proceeds as follows. Section 2 develops the algebraic foundation: the cosmic causal diamond’s algebra of observables $\mathcal{A}(\mathcal{D})$, the modular Hamiltonian K , and the holographic information-processing rate $\gamma(z)$. Section 3 develops the structural account of time: γ as the foundational rate of state precipitation, atomic clocks slaved to γ , the impossibility of detecting γ -variation from inside a single causal diamond. Section 4 develops the structural account of scale: each cycle’s measurement frame is internally consistent and externally incommensurable, no absolute meter exists, the previous universe’s “size” is not a meaningful question. Section 5 develops cyclicity: the QTEP partition between coherent and decoherent entropy drives the $H \rightarrow 0$ asymptote toward a maximum-coherence configuration that is structurally unstable to decoherence cascades, with cycle transitions instantaneous to internal observers. Section 6 develops the boundaries of physics: which questions are answerable within the framework, which dissolve as frame-dependent, and how the boundaries of measurement structure the boundaries of meaningful cosmological inquiry. Section 7 examines traditional cosmological puzzles in light of these commitments. Section 8 concludes.

This is a foundational paper. It contains no new equations beyond those developed in [7,8,9]; it makes no new empirical predictions; it reports no new measurements. The contribution is to articulate, on its own terms, what those predictions and measurements commit us to about the nature of time, space, and scale. The argument is structural: given the algebraic foundation, the consequences for cosmological measurement and meaning follow.

2 The algebraic foundation

2.1 The cosmic causal diamond and its algebra of observables

The cosmic causal diamond $\mathcal{D}$ is the spacetime region $\mathcal{V}(p, q) = J^+(p) \cap J^-(q)$ bounded by two null sheets meeting at a 2-surface σ of maximal area — the holographic screen [10,11,12]. For an observer at the present epoch, $\mathcal{D}$ extends from their causal past at the Big Bang to their causal future at the de Sitter horizon, with σ the screen on which information from the bulk is encoded.

The algebra $\mathcal{A}(\mathcal{D})$ of quantum observables supported on $\mathcal{D}$ carries an intrinsic modular structure [13,14,15,16]. By the Bisognano-Wichmann theorem [13] and its extensions to bounded regions [14,15], $\mathcal{A}(\mathcal{D})$ is a Type III₁ von Neumann algebra; its Tomita-Takesaki modular flow generates an automorphism group whose generator K — the modular Hamiltonian — defines a thermodynamic notion of horizon-region energy and entropy. The state on $\mathcal{A}(\mathcal{D})$ is the modular vacuum, KMS at the modular temperature $T_{\text{mod}} = \hbar H/(2\pi)$ — the Gibbons-Hawking temperature of the cosmic horizon [17].

H- Λ CDM’s algebraic-primary stance treats $\mathcal{A}(\mathcal{D})$ as foundational. The modular state on $\mathcal{A}(\mathcal{D})$ is the foundational object; the Lorentzian metric, spacetime manifold, and matter content are derived from the modular state through the holographic projection developed in [7,8,9]. This inverts the standard direction of cosmological theorizing, which begins with a fixed Lorentzian background and adds quantum content as fields on that background. Within H- Λ CDM, the manifold is not a primitive — it is the Lorentzian representation of the modular state, recovered as an accurate description in the limit where the effects of holographic projection become negligible (where γ -variation across the relevant scales is small and the projection’s higher-order corrections fall below observational precision). Standard cosmology is the framework’s leading-order behavior in this limit; the cosmological tensions developed in [7,8,9] are the residue where that limit is not exact.

2.2 The modular Hamiltonian and the Bekenstein-Hawking entropy

The modular Hamiltonian on $\mathcal{A}(\mathcal{D})$ takes the form

$$K = 2\pi \int_{\Sigma} T_{\mu\nu}^{\text{info}} \xi^{\mu} d\Sigma^{\nu}, \quad (1)$$

where Σ is a Cauchy surface for $\mathcal{D}$, ξ^{μ} is the conformal Killing vector preserving the diamond and degenerating on its boundary, and $T_{\mu\nu}^{\text{info}}$ is the information stress-energy tensor of the modular state on $\mathcal{A}(\mathcal{D})$ [9]. The von Neumann entropy of the modular state evaluates to

$$S_{\text{max}} = \frac{\mathcal{A}}{4G\hbar} = \frac{\pi c^5}{G\hbar H^2}, \quad (2)$$

the Bekenstein-Hawking area law for the diamond [18,19]. This is a *derived* property of the modular state rather than an axiom imposed on the Lorentzian geometry; the holographic principle [10,11,12] is recovered as a structural consequence of taking the modular foundation seriously.

The number of holographic degrees of freedom on the cosmic horizon is $N_P = \mathcal{A}/\ell_{\text{Pl}}^2 \approx 9 \times 10^{122}$ at the present epoch — the number of Planck-area cells on the screen. This counts the maximum information capacity of the observable universe, in the sense of [10,11,12]. The information stress-energy tensor $T_{\mu\nu}^{\text{info}}$, the modular Hamiltonian K , and the entropy S_{max} all derive from the same algebraic foundation; they are not independent inputs.

2.3 The holographic information-processing rate $\gamma(z)$

The Margolus-Levitin theorem [20] bounds the rate at which a quantum system can transition between orthogonal states by $2E/(\pi\hbar)$, where E is the system’s mean energy. Applied to the cosmic horizon as a holographic computational substrate with energy $E_{\text{horizon}} = S_{\text{max}} \times T_{\text{mod}} = c^5/(2GH)$, this gives a maximum information-processing rate that, when expressed per logarithmic unit of horizon information capacity, takes the form

$$\gamma(z) = \frac{H(z)}{\ln N_P(z)}. \quad (3)$$

This rate is the holographic information-processing rate developed in [7,8]: the rate at which the cosmic horizon as a holographic screen processes information about the bulk content it encodes. At the present epoch, $\gamma(0) \approx H_0/281.7 \approx 7.7 \times 10^{-21} \text{ s}^{-1}$, well below standard cosmological frequencies. At recombination $z \approx 1100$, $\gamma \approx 8 \times 10^{-16} \text{ s}^{-1}$, $\sim 10^5$ times faster than today. The rate changes in proportion to the cosmic horizon's expansion: as the horizon's proper area increases (and hence N_P grows and H decreases), γ shrinks. The evolution is structural — tied to the horizon's information capacity through N_P — rather than temporal in any cycle-independent sense.

The rate γ enters cosmological observables through the modular flow on $\mathcal{A}(\mathcal{D})$. Under modular flow e^{-iKs} , the algebra evolves with parameter s , and the relationship between s and proper time τ along comoving worldlines is $ds = \gamma(z) d\tau$. The Lorentzian-projection metric $H(z)$ and the modular flow rate $\gamma(z)$ are coupled through the dimensionless ratio $\gamma/H \approx 1/281.7$, approximately constant across cosmic history because the logarithm of the holographic information capacity varies slowly. This near-constancy of γ/H across epochs is what makes it useful as a perturbative expansion parameter; the framework's predictions in [8,9] are computed at leading order in γ/H , with corrections at $O((\gamma/H)^2) \approx 10^{-5}$ negligible at current observational precision.

2.4 The Lorentzian projection

Standard cosmological observables — the Hubble parameter $H(z)$, the matter density $\rho_m(z)$, the CMB power spectrum, the BAO scale, distance-redshift relations — are projections of the modular state's structure onto the Lorentzian representation accessible to embedded observers. The projection is exact to leading order in γ/H and acquires corrections at $O(\gamma/H)$ and higher that are responsible for the cosmological-scale observable signatures developed in [7,8,9].

The Lorentzian metric, in this picture, is not a fixed background on which physics happens; it is a derived structure from the modular state, recovered as an accurate description in the limit where γ -variation across the relevant scales is negligible. Where γ -variation is not negligible — across cosmic history at large redshift separations — the projection acquires higher-order corrections that produce the cosmological tensions as their observational signature. The tensions are not failures of measurement; they are the framework's structural prediction that the Lorentzian projection's leading-order behavior cannot perfectly track the modular state across all scales.

This is the algebraic-primary commitment in compressed form: $\mathcal{A}(\mathcal{D})$ and its modular state are foundational; the Lorentzian manifold and its observables are derived; the derivation is exact only at fixed γ and acquires corrections wherever γ varies. Cosmological observation is therefore intrinsically scale-relative — different scales sample different γ values, hence different aspects of the modular state, hence different projections of the underlying algebraic structure into the observer's Lorentzian frame.

3 Time as embedded

3.1 The structural claim

Time, within H - Λ CDM, is not a property of an external manifold. It is the rate of state precipitation generated by modular flow on $\mathcal{A}(\mathcal{D})$, measured against an observer's embedded position in the algebra. Each observer's local time emerges from modular ticks accumulated within their own constitution — their atomic clocks, their measuring instruments, their physiological processes are all subsystems of $\mathcal{A}(\mathcal{D})$ whose state changes are driven by the same modular flow that drives every other state change in the algebra. There is no privileged external clock; there is no absolute time against which the observer's time can be calibrated. Time is what modular flow produces in the embedded measurement frame.

This is a stronger claim than standard relativity. In Einstein’s relativity, observers in different reference frames disagree about durations because the metric is invariant under Lorentz transformations, but they all share a single underlying spacetime. The relativity is between *frames* on a common manifold. In H- Λ CDM, the relativity is between *observers in different causal diamonds* with different modular states and different γ values. The diamonds are the foundational structures; the manifold is what emerges in their Lorentzian projections.

3.2 The slaving of atomic clocks to γ

A hydrogen 21 cm transition involves $\hbar$ of action per oscillation. If $\hbar$ is the fundamental quantum of action and γ is the foundational rate of state change, then each oscillation involves a fixed number of modular ticks — the same number at every cosmological epoch. The atomic-second-to-modular-tick ratio is therefore constant across cosmic history; what varies with γ is the absolute rate at which modular ticks accumulate.

This has a direct consequence for what observers measure. An observer at recombination, embedded in the recombination-epoch causal diamond, measures hydrogen oscillating at 1420 MHz against their atomic-second clock. An observer today, embedded in the present causal diamond, measures the same 1420 MHz against their atomic-second clock. Both measurements are correct in their respective frames. But the recombination-epoch atomic second contains more modular ticks per unit of any external reference (if such a reference existed) than today’s atomic second does, because γ at recombination was 10^5 times larger than γ today. The atomic second has lengthened over cosmic history, and atomic clocks tick more slowly today than at recombination — but observers cannot detect this slowdown because their measurement instruments are themselves slaved to γ at their epoch.

The undetectability is structural, not technical. It is not that we lack precision to detect the slowdown; it is that any measurement of γ -variation requires comparison between epochs, and within a single epoch’s measurement frame, γ is the rate against which all measurements are calibrated, hence not measurable. To detect γ -variation we must observe across epochs — the Hubble tension is exactly this kind of cross-epoch comparison, integrating γ along the null geodesic from recombination to today and measuring the residue of γ -variation as the H_{local} vs H_{CMB} differential.

3.3 What it means to say “time slows”

The statement “ γ shrinks as the cosmic horizon expands” is therefore meaningful within H- Λ CDM but does not translate to “time slows as the universe ages” in the operational sense familiar from special relativity. The framework rejects “aging” as an absolute time coordinate against which γ could be said to evolve. What γ tracks is the horizon’s information capacity through N_P , which grows as the horizon’s proper area grows; γ shrinks structurally because the denominator $\ln N_P$ grows faster than the numerator H . The evolution is tied to horizon expansion, not to the passage of time in any cycle-independent sense. Observers do not experience their seconds as longer than past observers experienced theirs; observers always experience their seconds as seconds, because their experience is itself slaved to γ . The structural claim is that the absolute rate at which modular ticks accumulate per unit of horizon expansion has decreased across the cycle’s history — a fact accessible only through cross-epoch ratio observables.

The Hubble tension is the cleanest such observable. H_{local} and H_{CMB} measure expansion rates at different cosmic epochs, both calibrated against atomic clocks at those epochs. In standard Λ CDM they should give the same H_0 because the metric is uniform. In H- Λ CDM they differ by $(\gamma/H) \times \ln(d_{\text{CMB}}/d_{\text{local}})$, the integrated γ -variation across the logarithmic distance separation. The $6 \text{ km s}^{-1} \text{ Mpc}^{-1}$ residue between SH0ES and Planck is the empirical signature of γ varying

across 13.8 billion years of cosmic history.

Other cross-epoch observables encode γ -variation similarly. The cosmological-constant scale $\rho_\Lambda(0) = 5.84 \times 10^{-27} \text{ kg m}^{-3}$ identifies as the inverse holographic information capacity at the present epoch, $\rho_\Lambda/\rho_{\text{Pl}} \approx \Omega_\Lambda/N_P(0)$; the same construction at higher z would give a different inverse-capacity ratio because N_P depends on the epoch’s horizon area [9]. The BAO sound horizon $r_d = 150.71 \text{ Mpc}$ emerges from integrating γ along the photon-baryon plasma’s evolution from γ at very early times to γ at recombination [7]. Each observable is sensitive to γ at the epoch it samples, and consistency across observables requires a single $\gamma(z)$ function that the framework specifies algebraically.

3.4 The asymmetry of internal vs cross-epoch measurement

Internal measurement — what an observer at any single epoch measures with their local instruments — never reveals γ -variation. Atomic frequencies appear constant; physical constants appear constant; the laws of physics appear identical. This is the empirical content of the equivalence principle for cosmological epochs: any local laboratory experiment yields the same result at recombination as today, when measured with local atomic clocks at the relevant epoch. The Standard Model coupling constants, masses, $\hbar$, e , m_e , c — all are universal in this sense, because they are ratios of modular-flow processes that scale together with γ .

Cross-epoch measurement — comparison of observables sampled at different epochs through null-geodesic integration — does reveal γ -variation. The Hubble tension, the BAO scale, the cosmological-constant ratio, and (if it persists) the DESI evolving-DE preference are all such cross-epoch comparisons. The asymmetry between internal invariance and cross-epoch variation is not a measurement artifact; it is the structural content of the framework’s algebraic-primary commitment. Internally, observers cannot escape γ ; cross-epoch, γ is precisely what is observed.

4 Scale as cycle-local

4.1 The cycle-locality claim

Scale, within H- Λ CDM, is not an absolute property of physical structure. It is the ratio between a physical length and an observer’s local meter, with the local meter defined by the speed of light times the observer’s local atomic second. Since the atomic second is slaved to γ at the observer’s epoch, and γ varies with epoch, the meter varies with epoch. There is no cross-epoch absolute meter against which lengths can be calibrated. Each cycle’s measurement frame is internally consistent — within the cycle, all measurements share a common γ -defined ruler — and externally incommensurable with other cycles’ frames.

This dissolves several familiar cosmological questions. The “size of the universe at the big bang” is not a meaningful absolute quantity; it is a measurement-frame statement about the boundary of the current cycle’s null structure, expressed in the current cycle’s units. Asking what the universe was “really” the size of at the big bang presupposes a cross-cycle meter that does not exist within the framework. The traditional answer (the universe was smaller than a Planck length at very early times) is correct only as an intracycle statement: in our cycle’s units, the very early universe was small. It does not translate to any absolute claim about size.

Similarly, the question “what was the universe’s age before the big bang” presupposes a clock that survives cross-cycle, which the framework does not provide. The very early universe of our cycle is the precipitation event from the previous cycle’s asymptotic decoherence cascade (developed in Section 5). The previous cycle’s “duration” is measured in its observers’ units, not ours, and there is no transformation rule that converts between them. The previous cycle

existed; its history was real for its observers; but its scale is not accessible from within our cycle’s measurement frame.

4.2 The structure of intracycle measurement

What can be measured within a cycle is the structure produced by γ -variation across that cycle’s history. The Hubble tension reveals γ varying across our cycle from recombination to today. The BAO horizon reveals γ varying across our cycle from very early times to recombination. The cosmological-constant scale reveals γ at the present epoch. The DESI dark-energy data, if confirmed as evolving, would reveal additional γ -evolution across $z \sim 0.5$. All of these are intracycle observables, each calibrated against the cycle’s local atomic standards, each contributing to a consistent reconstruction of $\gamma(z)$ within the cycle.

The observables consistent with our cycle’s $\gamma(z)$ function are the framework’s predictions. They are sharp because the algebraic structure of $\mathcal{A}(\mathcal{D})$ determines $\gamma(z)$ uniquely given $H(z)$ and $N_P(z)$; they are testable because they convert to specific quantitative claims about ratios of observables ($H_{\text{local}}/H_{\text{CMB}}$, r_d/d_M , $\rho_\Lambda/\rho_{\text{crit}}$). They are falsifiable because the framework cannot accommodate alternative $\gamma(z)$ functions to fit data; γ is structurally determined, not parametrically free.

What cannot be measured is the absolute scale of any cycle in cycle-independent units. The universe’s size, age, energy content, total entropy — all of these can be reported in our cycle’s units, but the reports do not transfer to other cycles. A future observer in the next cycle would report similar-form numbers for their universe, derived through the same algebraic structure, but the underlying scales would be incommensurable with ours. This is not a deficiency of measurement; it is the framework’s structural claim that absolute scale is not a meaningful concept.

4.3 The relativity of “expansion”

Within standard Λ CDM, the universe’s expansion is a property of the metric. The scale factor $a(t)$ grows over cosmic time, and physical distances stretch with $a(t)$. This is a frame-independent claim within the framework, in the sense that all observers agree the universe is expanding; the disagreement (under cosmological-perturbation theory, etc.) is over rates and details, not over the basic fact.

Within H- Λ CDM, the structural meaning of “expansion” is reconsidered. The cosmic horizon’s area increases in our local meters; this is what we call “expansion.” But the meter is defined by γ , and γ shrinks as the horizon’s proper area grows. The horizon’s area in absolute units (if such existed) is changing differently from how we measure it. What we observe as accelerating expansion in the late universe is the combined effect of the horizon area genuinely increasing and our meters shrinking faster as γ shrinks faster. Both contribute; the structural content is in their combined effect on cross-epoch ratios.

This does not mean the universe “isn’t really expanding.” It means “expansion” is itself a frame-relative concept. The horizon-area-vs-meter ratio that we measure as expansion is real and consistent within our frame. But it is not a fact about an absolute structure independent of measurement; it is a fact about what the modular flow produces in our embedded observation. A future observer in the next cycle would measure their cycle’s expansion in their meters and get a similar functional form; the underlying scale would be different but the structural pattern would repeat.

5 QTEP and cyclicity

5.1 The Quantum-Thermodynamic Entropy Partition

The modular state on $\mathcal{A}(\mathcal{D})$ carries entropy that decomposes into two thermodynamically distinct components: coherent entropy, stored in superposition over basis states; and decoherent entropy, stored in classicalized definite outcomes. The Quantum-Thermodynamic Entropy Partition (QTEP) [21] specifies the structural ratio between these components, with the decoherent fraction representing the irreducible thermodynamic floor produced by the obit — the fundamental unit of irreversibility, equal to 1 nat — and the coherent fraction representing what the modular state stores in superposition before any decoherence event.

The partition is not free. It is determined by the algebraic structure of $\mathcal{A}(\mathcal{D})$ and the modular state’s KMS condition, with the obit emerging as the structural minimum of decoherent entropy required for a state to be thermodynamically distinguishable from its coherent superposition. A state with zero decoherent entropy is not a meaningful physical state in the framework; the obit serves as the floor below which classical reality cannot be sustained.

QTEP has two consequences for cosmological structure that bear directly on the cyclicity argument. First, no physical state can be at absolute zero temperature, because absolute zero would require zero modular flow and hence zero decoherence rate, which violates the obit floor. The modular state at any epoch carries finite γ and finite decoherence rate. Second, no physical state can be fully decoherent, because full decoherence would eliminate the coherent reservoir that sources the modular flow’s renewal of classical structure. Reality maintains a finite ratio of coherent to decoherent entropy at every epoch, with the ratio shifting as γ evolves but never reaching either extreme.

5.2 The $H \rightarrow 0$ asymptote under QTEP

The de Sitter asymptote at $H_\infty \approx \sqrt{\Lambda/3}$ is the standard Λ CDM endpoint: matter dilutes to zero density, the cosmic horizon stabilizes at fixed area, and the universe approaches pure thermal equilibrium with the Gibbons-Hawking horizon. Within H- Λ CDM, this asymptote is not the terminus. The framework’s structure is consistent with H continuing to shrink past H_∞ in regimes where standard Λ CDM does not predict further evolution — specifically, regimes where the cosmological-constant component itself responds to γ -evolution at higher orders than the present analysis.

In the deep $H \rightarrow 0$ regime — denote it $H_{-10^{21}}$ for concreteness, far below H_∞ — γ has shrunk enormously. The modular flow rate has crashed to vanishingly small values measured against any external reference (which does not exist) or against the residual atomic-clock rate (which has crashed proportionally). The cosmic horizon has grown enormously in the local meter; N_P has grown to astronomical values; $\ln N_P$ has grown to dominate γ ’s denominator. The modular state on $\mathcal{A}(\mathcal{D})$ approaches the QTEP-coherent endpoint: maximum coherent entropy, minimum decoherent entropy consistent with the obit floor, lowest accessible thermodynamic temperature consistent with QTEP.

This is not heat death in the standard thermodynamic sense. Heat death is an equilibrium state with maximum entropy; the $H \rightarrow 0$ endpoint is a non-equilibrium state with maximum *coherent* entropy and a small but finite decoherent residue. The system is structurally far from the QTEP-preferred ratio, which is balanced at intermediate temperatures, not at the extreme cold of maximum coherence. The $H \rightarrow 0$ endpoint is metastable in QTEP terms — it has the entropy capacity to evolve toward the partition’s preferred ratio, requiring only a decoherence seed to initiate the relaxation.

5.3 The decoherence cascade

Maximum coherence is structurally unstable to decoherence under QTEP. Spontaneous decoherence events — quantum fluctuations in the coherent reservoir that classicalize into definite outcomes — are not Boltzmann-suppressed in the usual way, because the coherent reservoir has uniform low temperature throughout, with no temperature gradient to suppress fluctuations. Each spontaneous decoherence event releases the energy gradient between coherent and decoherent entropy as new classical structure, increasing the local information density and hence increasing γ at that locus.

A single decoherence seed in the maximum-coherence reservoir is structurally amplifying. Its precipitation increases γ locally; the increased γ accelerates further decoherence in its causal vicinity; the accelerating decoherence cascades through the reservoir as a propagating front. The cascade releases the coherent reservoir’s stored entropy as classical structure at increasing rates, with γ building from negligible asymptotic values back to high cosmological values across the cascade’s duration.

To external description (which does not exist within the framework) the cascade duration would be enormous — the rate of cascade propagation is itself γ -limited, and γ in the asymptote is vanishingly small. To internal observers inside the cascade, the cascade is the very fabric of cosmic history at the new cycle’s earliest epochs. They experience high γ , rapid precipitation, classical structure emerging from their own observational frame’s “initial conditions.” Their hot, dense early universe is the cascade’s leading edge from their perspective, and their cosmic time begins at the cascade’s first decoherence event — what we call the big bang of the new cycle.

5.4 Cycle transitions are immediate to internal observers

The asymmetry between external and internal description is structural under embedded measurement. From any internal observer’s frame, the cycle they are in is the only cycle accessible, and its history begins at their cycle’s earliest decoherent event. The previous cycle’s asymptotic coherent reservoir is on the wrong side of their null structure — accessible only as the boundary condition that sourced their cycle’s precipitation, not as a temporally prior state in their measurement frame.

This dissolves the question “how long does the cycle transition take?” The transition has no duration in any internal frame. The previous cycle’s last detectable observer (assuming any can persist into the deep $H \rightarrow 0$ regime, which the framework does not guarantee) experiences their cosmic history as approaching the asymptote with γ shrinking; eventually their measurement frame can no longer sustain measurement, and from their frame the next cycle never begins. The next cycle’s first detectable observer experiences their cosmic history as starting at the cascade’s first decoherent event; from their frame, the previous cycle’s asymptote is the boundary condition that precedes their accessible spacetime.

The transition is real; it is what links the cycles together algebraically. But it is not a temporally extended process from any internal frame. Each cycle is its own complete cosmological history, with its own beginning, middle, and asymptote, in its own units. The cycles are linked structurally through QTEP precipitation but not through any internal observer’s continuous experience.

5.5 Why this is structurally cyclic, not anthropic

The QTEP-driven cyclicity argument is structural rather than anthropic. It does not require that any internal observer survive the cycle transition or that subsequent cycles support observers; it requires only that the algebraic structure of $\mathcal{A}(\mathcal{D})$ at the $H \rightarrow 0$ asymptote is unstable to decoherence cascades, and that the cascades produce new modular-flow regimes algebraically

equivalent to the original. The argument is from the partition’s structural commitments to the asymptote’s instability to the cascade’s algebraic completeness, not from observer requirements at any stage.

This distinguishes the framework’s cyclicity from anthropic multiverse arguments [22], conformal cyclic cosmology [23], or ekpyrotic models [24]. The framework does not select among possible cycles based on observer presence; all cycles are structurally identical (algebraically equivalent modular structures with their own scales and observers). The framework does not require an external conformal rescaling to map the asymptote to a new beginning; the cascade is the rescaling, performed internally by QTEP relaxation. The framework does not require brane collisions or extra-dimensional dynamics; the cyclicity is intrinsic to the cosmic causal diamond’s algebraic structure.

The cycles are not “different universes” in the multiverse sense. They are sequential algebraic instantiations of the same structural pattern — $\mathcal{A}(\mathcal{D})$, modular flow, QTEP partition, γ -evolution, asymptotic coherence, decoherence cascade — with each instantiation carrying its own scale and its own observers but the same structural commitments. The framework is cyclic because the $H \rightarrow 0$ asymptote is QTEP-unstable, full stop. The cyclicity is a structural prediction, not a model-building choice.

6 The boundaries of physics

6.1 What is measurable

Within Λ -CDM, measurement is intrinsically intracycle and intrinsically scale-relative. What is measurable is what can be expressed as a ratio of modular-flow processes within a single observer’s causal diamond. The Hubble parameter measured against the atomic second is measurable. The cosmological-constant density measured against the critical density is measurable. The BAO horizon measured against the comoving distance scale is measurable. Each of these is a ratio observable that carries empirical content within the cycle that produces it.

The framework’s parameter-free predictions are claims about specific values these ratio observables should take. The BAO sound horizon prediction $r_d = 150.71$ Mpc [7] is a claim about what our cycle’s atomic standards measure when the photon-baryon plasma’s history is integrated. The Hubble local-vs-CMB prediction $H_{\text{local}}/H_{\text{CMB}} = 1 + (\gamma/H) \ln(d_{\text{CMB}}/d_{\text{local}})$ [8] is a claim about our cycle’s H_0 measurements at different anchor distances. The cosmological-constant prediction $\rho_\Lambda/\rho_{\text{crit}} = \Omega_\Lambda$ at $z = 0$ [9] is a claim about our cycle’s vacuum-energy measurements relative to the critical density.

Each prediction is sharp, falsifiable, and parameter-free within our cycle. The empirical record so far supports all three: BAO observations match the framework’s prediction within current precision [7]; conservative-TRGB measurements match the Hubble prediction across photometric systems [8]; the cosmological-constant scale matches the inverse-information-capacity identification [9]. The framework’s predictions are within-cycle observables that the framework’s algebraic structure determines uniquely from H_0 , $\hbar$, G , and the algebra’s combinatorial structure.

6.2 What is not measurable

What is not measurable is anything that requires cross-cycle calibration. The absolute size of the universe at the big bang is not measurable — there is no cross-cycle meter to express it in. The total energy of the observable universe is not measurable in any cycle-independent sense — there is no cross-cycle joule. The duration of the previous cycle is not measurable — there is no cross-cycle second. These are not failures of measurement precision; they are structural

consequences of intracycle measurement being the only kind of measurement the framework supports.

The traditional cosmological literature treats these as questions awaiting better data or better theory. Within H- Λ CDM they are dissolved as frame-dependent artifacts. Asking how big the universe was “really” at the big bang is asking for a transformation between our cycle’s meters and an absolute meter that does not exist. Asking what came “before” the big bang is asking for a temporal extension of our cycle’s clock into a regime where our cycle’s clock does not apply. These are not deep questions about hidden physics; they are mistakes about what cosmological measurement can do.

This narrows the legitimate scope of cosmological inquiry. The framework forbids certain questions as meaningless within its structure. This is not a rhetorical move; it is a structural commitment that follows from algebraic primacy. If $\mathcal{A}(\mathcal{D})$ is foundational and cosmological observables are projections from it, then questions that presuppose absolute structure beyond what the algebra provides are not poorly-formed empirical claims — they are not empirical claims at all. The framework does not simply lack answers to such questions; it asserts the questions do not have answers in the sense the asker intends.

6.3 The philosophical implication: cosmology as relational

The structural commitments developed in Sections 3–5 amount to a relational cosmology. Time is the rate of state precipitation relative to the embedded observer. Scale is the ratio of physical lengths to the observer’s local meter. Cyclicity is the algebraic structure linking cycles through QTEP relaxation. Each of these is relational rather than absolute: each depends on the observer’s embedding in $\mathcal{A}(\mathcal{D})$, and none is meaningful as an absolute property of an external manifold.

This places H- Λ CDM in conversation with relational physics programs developed for other reasons. Rovelli’s relational quantum mechanics [25,26] treats quantum states as relational rather than absolute. Smolin’s argument for a real ontology of time [27,28] develops time as fundamental rather than emergent. Penrose’s conformal cyclic cosmology [23] uses conformal rescaling to link cycles. Bousso’s holographic-bound work [12] treats the holographic principle as foundational. Each of these programs makes structural commitments adjacent to the H- Λ CDM framework’s, sometimes overlapping and sometimes diverging.

The H- Λ CDM framework’s distinctive contribution is that its relational commitments are structurally derived from the algebraic foundation rather than philosophically motivated. We did not start with a desire for a relational cosmology and then build the framework to support it; we started with the algebraic structure of $\mathcal{A}(\mathcal{D})$ and the relational consequences emerged as structural implications. The framework’s empirical successes [7,8,9] validate the algebraic foundation, and the relational commitments developed in this paper are what that foundation requires once articulated.

This is a stronger position than philosophical motivation. It is a claim that physics, properly understood through its algebraic foundation, is relational. The traditional Lorentzian-manifold formulation that treats spacetime as primary is the framework’s leading-order projection, accurate where γ -variation is negligible across the relevant scales. Where γ -variation is not negligible — across the cosmological history that produces the Hubble tension and the cosmological-constant scale — the projection’s higher-order corrections become observable, and what those corrections describe is the relational structure that the framework’s algebraic foundation has been telling us about all along.

7 Reframing traditional cosmological puzzles

7.1 The size of the universe at the big bang

In standard cosmology, the early universe was small in absolute terms. A common pedagogical claim is that the observable universe at recombination occupied a sphere of radius ~ 40 Mpc, and at very early times the entire observable universe was smaller than a proton. These are intuitive statements about absolute scale.

Within H- Λ CDM these statements require reframing. The early universe was small *in our current cycle's meters*. The meter is defined by our atomic second times the speed of light, with the atomic second slaved to γ at the present epoch. Projecting our meter back to recombination and reporting that the universe was 40 Mpc across at that epoch is a valid cross-epoch ratio observable: we are measuring the recombination-epoch horizon size in our atomic standards transported via null geodesics to today. The number is correct as such.

But the statement “the universe was small” suggests that smallness is an absolute property — that there is some sense in which the universe at recombination was objectively smaller than the universe today by some universal factor. The framework forbids this stronger interpretation. The recombination universe was the size that observers at recombination, with their atomic clocks and meters, would have measured it to be — and they would have measured it as a normal-sized universe in their frame, not a small one. Smallness is our retrospective measurement, transported through our γ -evolved meter to a regime where their γ was different.

The honest statement is: relative to our cycle's meters, the recombination universe was small. This is empirically correct and operationally meaningful. It is not a claim about absolute smallness, because absolute size is not a meaningful concept.

The same frame-dependence applies to $t = 0$ itself. From inside our cycle, looking back along our null structure to where it terminates, $t = 0$ appears as a degenerate causal diamond — the boundary at which our cycle's algebra first becomes nontrivial, with the diamond's proper extent at its measurement-frame minimum. Standard cosmology describes this as the initial singularity, the universe collapsing to a point. Within H- Λ CDM the picture is sharper: $t = 0$ is the cascade's leading edge from the previous cycle's asymptotic coherent reservoir, and our null structure terminates at the cascade's first decoherent event because there is no earlier decoherent event in our cycle's history. We measure this as a degenerate diamond because we are calibrating against our cycle's expanded N_P , and the leading-edge state has minimal N_P in our measurement frame.

But the degeneracy is a measurement artifact, not a structural fact about the cascade transition. From the previous cycle's frame (now inaccessible), the cascade's leading edge was precipitating from a saturated coherent reservoir with vast N_P — the opposite of degenerate. The same structural state appears as “tiny degenerate boundary” in our cycle's frame and as “saturated information-rich asymptote” in the previous cycle's frame, with no cycle-independent description that picks one over the other. Saying “the universe began as a degenerate diamond” is therefore not a claim about the cycle's intrinsic structure; it is a claim about how the cycle's intrinsic structure appears from inside our cycle's measurement frame. The cross-cycle algebraic structure that links the cycles has no privileged frame from which the cascade transition can be described as “really” degenerate or “really” saturated. Both descriptions are correct in their respective frames, and the framework does not privilege either.

This deepens the dissolution of “the universe was small at the big bang” as a meaningful absolute claim. Even the apparent degeneracy of $t = 0$ is intracycle — what we infer from our null structure terminating at the cascade boundary, calibrated against our cycle's atomic standards. The cycle-locality of measurement applies not just to size and duration but to the very structural status of the cycle's beginning. The frame-dependent picture is the only picture available; there

is no frame-independent answer to “what was happening at $t = 0$ ” beyond the algebraic structure of the cascade transition itself, which links the cycles without privileging either.

7.2 What came before the big bang

The traditional question presupposes a temporal extension of cosmic history into a regime preceding our cycle’s earliest decoherent event. Within H- Λ CDM, this is the previous cycle’s asymptotic coherent reservoir — the maximum-coherence configuration that QTEP-cascade-precipitated into our cycle’s beginning. The previous cycle’s history is real in the sense that it produced the algebraic boundary conditions our cycle inherited. But it is not accessible as a temporally prior state in our measurement frame.

The cascade transition has no duration from any internal observer’s frame. From the previous cycle’s last observers, the cascade is something they cannot measure because their measurement frame fails as γ approaches the asymptote. From our cycle’s first observers, the cascade is the boundary condition before which our null structure does not extend. The transition is real algebraically but not temporally extended in any internal frame.

Asking what came “before” the big bang is asking for a temporal extension that does not exist in any frame. The previous cycle’s history is not “before” our cycle’s history in any meaningful internal-clock sense; it is on the other side of the algebraic boundary that links the cycles. The traditional question is reframed within H- Λ CDM as: what is the algebraic boundary condition our cycle’s modular flow inherited from the previous cycle’s QTEP cascade? That question has structural answers (the cascade structure determines the early-cycle conditions); it does not have a temporal answer of the kind the traditional question asks.

7.3 The fine-tuning of initial conditions

Standard cosmology faces the question of why our universe’s initial conditions appear fine-tuned for structure formation. The flatness problem, the horizon problem, and the cosmological-constant problem are traditional formulations of this puzzle. Inflationary cosmology [29,30] proposes solutions involving exponential expansion that smooths and flattens initial conditions; multiverse arguments [22] propose anthropic selection across a vast ensemble of universes.

Within H- Λ CDM, the apparent fine-tuning is reframed as the structural consequence of QTEP precipitation. The cascade that initiates each cycle starts from the previous cycle’s maximum-coherence asymptote; the cascade’s structure is determined by QTEP relaxation toward the partition’s preferred ratio; the cascade’s structural pattern is the same across all cycles. Each cycle inherits initial conditions that match the QTEP cascade’s structural specifications, not because of fine-tuning or selection, but because the cascade structurally produces those conditions.

Flatness is a structural prediction within H- Λ CDM (Section 2 of [9]; Section 4 of this paper): $\rho_{\text{info}}(z) = \rho_{\text{crit}}(z)$ at every epoch follows from the modular state’s energy density divided by the Hubble volume. The flatness is not fine-tuned; it is structural. The cosmological-constant scale is similarly structural: $\rho_{\Lambda}/\rho_{\text{Pl}} \approx \Omega_{\Lambda}/N_P(0)$, the inverse holographic information capacity. The cosmological-constant problem is dissolved as a counting argument rather than a fine-tuning. The horizon problem is dissolved by recognizing that γ at the early universe was much larger than today, and the horizon-sized correlations of CMB observations reflect γ -evolution rather than acausal fine-tuning.

What inflation accomplishes phenomenologically — flat universe, smooth initial conditions, scale-invariant primordial spectrum — H- Λ CDM accomplishes structurally without invoking exponential expansion of a scalar field. The framework’s algebraic foundation requires these features as direct consequences. Whether inflation also occurs as a separate physical mechanism

is a question for future work; the framework does not require it for the structural results that inflation was originally proposed to address.

7.4 The arrow of time and irreversibility

The thermodynamic arrow of time — entropy increases over cosmic history — is a structural prediction within H- Λ CDM. The QTEP partition’s drive toward the preferred coherent-decoherent ratio is the algebraic source of irreversibility, with the orbit emerging as the fundamental unit of irreversibility (1 nat per decoherence event). Cosmic time evolution is the modular flow’s progressive precipitation of coherent entropy into decoherent entropy, accumulating irreversible classical structure as cosmic history progresses.

This grounds the arrow of time in the algebraic foundation rather than in initial-condition specification. Standard cosmology faces the question of why initial conditions had low entropy, with various proposals (Penrose’s Weyl curvature hypothesis [31], for instance) addressing why the early universe had the structural properties it did. Within H- Λ CDM, the early universe’s initial conditions are the QTEP cascade’s leading edge, with high γ and rapid decoherence rate. The “low entropy” of the early universe in the standard formulation translates to “high coherent reservoir, rapid decoherence cascade” in the framework — a structural state of the cascade rather than a fine-tuned initial condition.

The cosmological arrow of time is therefore tied to the cycle’s QTEP relaxation. Each cycle starts at high γ and high coherent entropy, evolves through progressive decoherence at decreasing γ , and asymptotes to maximum coherence at low γ . The arrow points from cascade-onset toward asymptote; it does not point in some external absolute direction. The arrow is intrinsic to the cycle’s QTEP structure, not added as a separate boundary condition.

8 Conclusion

H- Λ CDM, when its algebraic foundation is taken seriously rather than treated as a tool for computing observables on a fixed background, requires a relational cosmology. Time is the rate of state precipitation relative to embedded observers, with γ varying across cosmic history in ways internal observers cannot detect but cross-epoch ratio observables can measure. Scale is intrinsically cycle-local, with no absolute meter linking cycles. Cyclicity follows structurally from the QTEP partition’s instability at the $H \rightarrow 0$ asymptote, with cycle transitions instantaneous to internal observers regardless of external duration. The boundaries of physics are accordingly the boundaries of intracycle measurement — questions that require cross-cycle calibration are dissolved as frame-dependent rather than answered.

The framework’s empirical successes — the BAO sound horizon prediction, the Hubble tension resolution, the cosmological-constant scale identification, the parameter-free $w(z)$ prediction — anchor these structural commitments. Three independent cosmological observables, three quantitative successes from a single algebraic foundation. The structural commitments are what that foundation requires once articulated; the empirical successes validate the foundation; the foundation in turn validates the structural commitments as the correct way to understand what the empirical successes are telling us.

This relational cosmology is not philosophically motivated; it is structurally derived. We did not begin with a preference for relationalism and build the framework around it; we began with the algebraic foundation and the relational commitments emerged as consequences. The traditional Lorentzian-manifold formulation of cosmology is the framework’s leading-order projection, recovered as an accurate description in the limit where the effects of holographic projection become negligible. The cosmological tensions of the last decade — Hubble, BAO, dark

energy, structure growth — are precisely the regimes where this limit is not exact at observable precision, and they are calling for the relational reformulation that the framework provides.

The traditional cosmological puzzles examined in Section 7 — the size of the universe at the big bang, what came before, the apparent fine-tuning of initial conditions, the arrow of time — are dissolved or reframed under the relational view. Each turns out to presuppose absolute structure that the framework does not support. The questions, properly understood, ask things that cannot be answered because they presuppose frames the framework forbids. This is not skepticism about cosmology; it is precision about what cosmological measurement can and cannot do.

The boundaries of physics are not where measurement precision runs out. They are where the algebraic structure ceases to support cross-frame comparison. Within the cycle, physics is precise, predictive, falsifiable. Across cycles, certain questions are simply not formulated correctly. Recognizing this distinction is not a retreat from cosmology’s empirical ambitions; it is a sharpening of those ambitions, focusing them on what can be measured and predicted, and dissolving what cannot.

Future work will develop the framework’s commitments at the deeper algebraic level — the explicit structure of QTEP precipitation cascades [21], the algebraic specification of cycle-to-cycle inheritance, the role of the orbit as the fundamental unit of irreversibility in the cascade dynamics, and the structural relationship between γ and the modular state’s Tomita-Takesaki structure on $\mathcal{A}(\mathcal{D})$. The empirical record will extend through DESI Y5, Euclid, Roman, JWST-anchored TRGB campaigns, DESI Year 3 BAO, and the cosmological-constant precision improvements that follow. Each of these tests is a test of γ -evolution within our cycle, and the framework’s algebraic structure means its predictions are fixed and cannot be adjusted to accommodate future data. If the predictions hold, the relational commitments developed in this paper are the correct way to understand what cosmology is telling us. If they fail, the framework’s algebraic foundation is wrong, and the relational commitments fall with it.

Either outcome will be informative. The empirical record will continue, and the framework’s structural commitments stand or fall with it.

Acknowledgments

[Placeholder]

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